

Local False Discovery Rates

Classic single-case hypothesis testing theory depended on the interpretation of tail-area statistics, that is, on p -values. The post-World War II boom in multiple testing continued to feature p -values, with their dominance extending into the large-scale testing era, as witnessed in Chapters 3 and 4. Even false discovery rate control, which strays far from significance testing and Type I error, is phrased in terms of p -values in its Benjamini–Hochberg formulation.

It is by necessity that tail areas are featured in single-case testing: one can sensibly interpret the probability that test statistic z exceeds 1.96, but $\text{Prob}\{z = 1.96\} = 0$ is uninformative. Large-scale testing, however, allows the possibility of *local* inference in which outcomes such as $z = 1.96$ are judged on their own terms and not with respect to the hypothetical possibility of more extreme results. This is the intent of local false discovery rates, a Bayesian idea implemented by empirical Bayes methods in large-scale testing situations.

5.1 Estimating the Local False Discovery Rate

As in Section 2.2, we begin with the Bayesian *two-groups* model, in which each of the N cases is either null or non-null, with prior probability π_0 or π_1 ,

$$\begin{aligned}\pi_0 &= \text{Pr}\{\text{null}\} & f_0(z) &= \text{null density} \\ \pi_1 &= \text{Pr}\{\text{non-null}\} & f_1(z) &= \text{non-null density},\end{aligned}\tag{5.1}$$

as diagrammed in Figure 2.3. Assumption (2.9), that π_0 is near 1, is not necessary as long as $f_0(z)$ is known; (2.9) won't be our assumption here, but in Chapter 6, where f_0 must be estimated, we will need π_0 near 1.

The local false discovery rate (2.14) is

$$\text{fdr}(z) = \text{Pr}\{\text{null}|z\} = \pi_0 f_0(z) / f(z)\tag{5.2}$$

where $f(z)$ is the *mixture density*

$$f(z) = \pi_0 f_0(z) + \pi_1 f_1(z). \quad (5.3)$$

Of the three quantities in (5.2), $f_0(z)$ is assumed known in this chapter while π_0 can be estimated as in Section 4.5 (or set equal to 1 with little harm if $\pi_0 \geq 0.90$). That leaves $f(z)$ to be estimated from the z -values $\mathbf{z} = (z_1, z_2, \dots, z_N)$, all of which, by definition of the mixture density, come from $f(z)$.

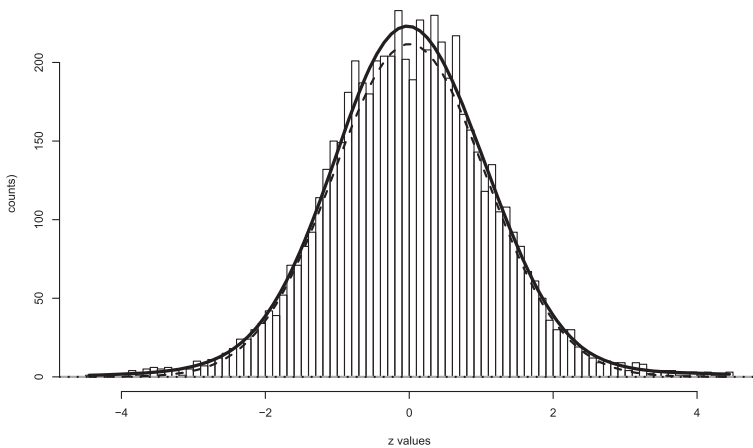


Figure 5.1a z -value histogram for the prostate data, $N = 6033$, Section 2.1. Heavy curve is estimate $\hat{f}(z)$ for mixture density $f(z)$, scaled to match histogram area. Dashed curve is scaled estimate $\hat{\pi}_0 f_0(z)$, where f_0 is the standard normal density (2.9).

Density estimation has a bad reputation in applied work, deservedly so given its pathological difficulties in discontinuous situations. Z -value densities, however, tend to be quite smooth, as discussed in Chapter 7. This can be seen in Figures 5.1a and 5.1b, which show z -value histograms for the prostate and DTI data sets. The heavy curves are estimates $\hat{f}(z)$ of $f(z)$ obtained as smooth fits to the heights of the histogram bars by a Poisson regression method described in Section 5.2.

The null probability π_0 in the numerator of (5.2) was estimated taking $\alpha_0 = 0.50$ in (4.48), that is, using the central 50% of the F_0 -distribution;

$$\hat{\pi}_0 = 0.932 \text{ (prostate data),} \quad \hat{\pi}_0 = 0.935 \text{ (DTI).} \quad (5.4)$$

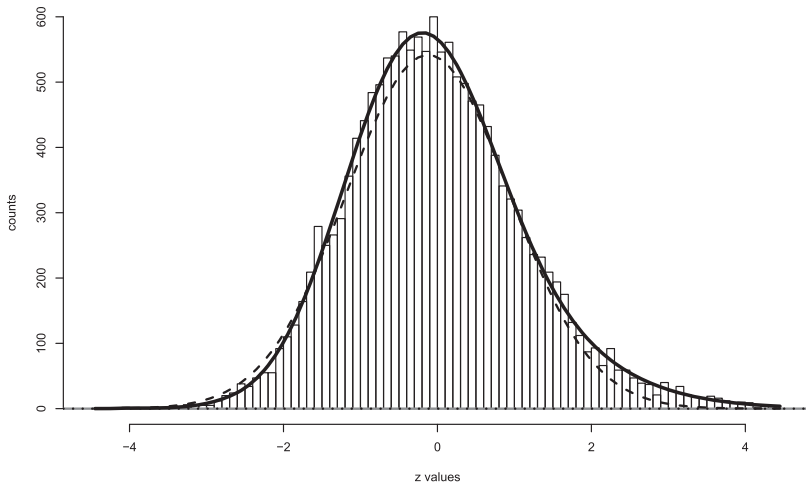


Figure 5.1b z -value histogram for the DTI data, $N = 15\,443$, Section 2.4; curves as in Figure 5.1a.

We can then estimate $\widehat{\text{fdr}}(z)$ by

$$\widehat{\text{fdr}}(z) = \hat{\pi}_0 f_0(z) / \hat{f}(z) \quad (5.5)$$

with $f_0(z)$ the $N(0, 1)$ density $\varphi(z)$ (2.9). The heavy curves in Figure 5.2 show $\widehat{\text{fdr}}(z)$. More precisely, they show $\min\{\widehat{\text{fdr}}(z), 1\}$. Equation (5.5) can exceed 1 because $\hat{\pi}_0$ isn't a perfect estimate of π_0 and, more seriously, because $\varphi(z)$ isn't a perfect estimate of f_0 . The latter can be seen in Figure 5.1b where $f_0(z)$ is shifted noticeably to the right of the histogram high point. Chapter 6 deals with much worse mismatches.

The estimate $\hat{f}(z)$ can be integrated to give a smoothed cdf estimate $\hat{F}(z)$, and then a smoothed version of the tail area false discovery rate,

$$\widehat{\text{Fdr}}(z) = \hat{\pi}_0 F_0(z) / \hat{F}(z), \quad (5.6)$$

here with $F_0(z) = \Phi(z)$, the standard normal cdf, or likewise a right-sided estimate as in (4.33). These are also shown in Figure 5.2. $\widehat{\text{Fdr}}(z)$ tends to be a little less variable than the non-parametric version $\overline{\text{Fdr}}(z) = \hat{\pi}_0 F_0(z) / \bar{F}(z)$ (4.26), especially in the extreme tails.

A conventional threshold for reporting “interesting” cases is

$$\widehat{\text{fdr}}(z_i) \leq 0.20. \quad (5.7)$$

This was achieved for $z \leq -3.37$ and $z \geq 3.34$ for the prostate data and $z \geq 3.05$ for the DTI data. $\widehat{\text{Fdr}}(z)$ was about half of $\widehat{\text{fdr}}(z)$ at these points,

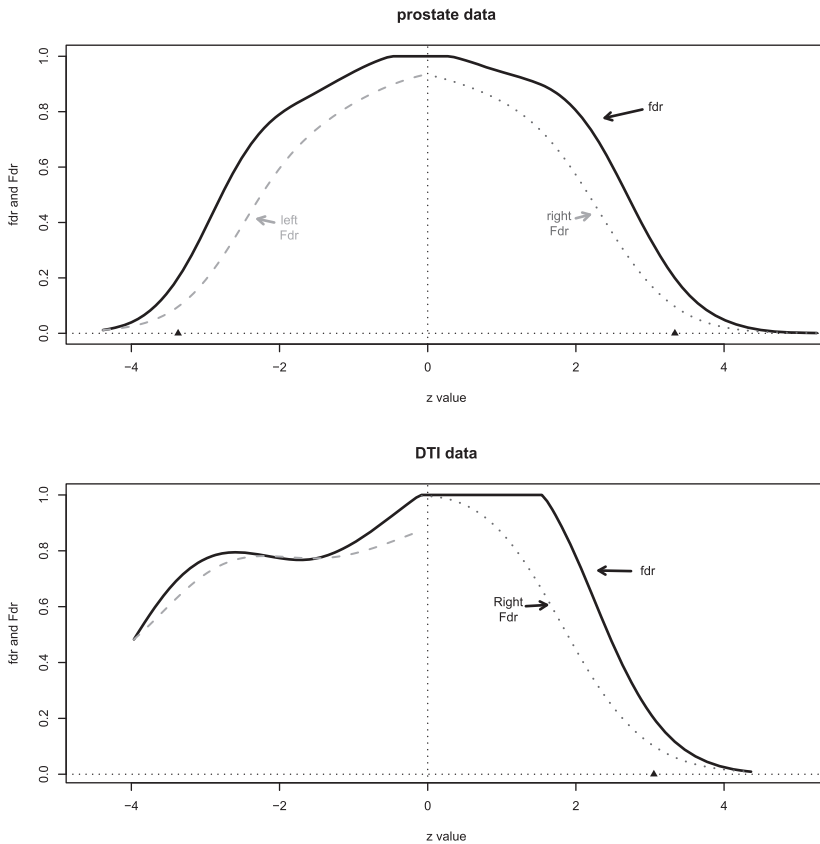


Figure 5.2 Heavy curves are estimated local false discovery rates $\widehat{fdr}(z)$ (5.5); dashed and dotted curves show estimated tail-area false discovery rates $\widehat{Fdr}(z)$ (5.6). The small triangles indicate values at which $\widehat{fdr}(z) = 0.20$; 27 genes in each tail had $\widehat{fdr}(z_i) \leq 0.20$ for the prostate data; 184 voxels, all in the right tail, had $\widehat{fdr}(z_i) \leq 0.20$ for the DTI data.

roughly following (2.17)–(2.19) with $\gamma = 2$. Criterion (5.7) was effectively about the same as the $q = 0.1$ Fdr threshold.

Exercise 5.1 Show that the criterion $\widehat{fdr}(z) \leq 0.2$ is equivalent to

$$\frac{f_1(z)}{f_0(z)} \geq 4 \frac{\pi_0}{\pi_1}. \quad (5.8)$$

Criterion (5.7) is based on the admittedly subjective grounds that reporting fdr values greater than 0.2 is dangerously prone to wasting investigators' resources. Like $\alpha = 0.05$ or $q = 0.1$, overuse of (5.7) is subject to Emerson's "foolish consistency" hobgoblin criticism, and perhaps no such criterion is necessary for interpreting a Bayesian posterior probability. My own experience nevertheless is that investigators often expect a definitive list of likely prospects, and (5.7) provides a reasonable way for the statistician to oblige.

If we assume $\pi_0 \geq 0.90$ as in (2.8), then (5.8) implies

$$\frac{f_1(z)}{f_0(z)} \geq 36. \quad (5.9)$$

The ratio $f_1(z)/f_0(z)$ is called the *Bayes factor* against the null hypothesis. Equation (5.9) requires a much stronger level of evidence against the null than in classical single-test practice: suppose we observe $z \sim \mathcal{N}(\mu, 1)$ and wish to test $H_0 : \mu = 0$ versus $\mu = 2.80$, a familiar scenario for power calculations since rejecting H_0 for $z \geq 1.96$ yields two-sided size 0.05 and power 0.80. Here the critical Bayes factor is only $f_{2.80}(1.96)/f_0(1.96) = 4.80$. We might rationalize (5.9) as necessary conservatism in guarding against multiple-testing fallacies. All of this relates to the question of power versus size, as discussed in Section 5.4.

5.2 Poisson Regression Estimates for $f(z)$

The smooth estimates $\hat{f}(z)$ for the mixture density $f(z)$, seen in Figure 5.1a, were obtained using maximum likelihood estimation (MLE) in flexible exponential family models.¹ As a first example, suppose f belongs to the J -parameter family

$$f(z) = \exp \left\{ \sum_{j=0}^J \beta_j z^j \right\}. \quad (5.10)$$

The constant β_0 in (5.10) is determined from $\beta = (\beta_1, \beta_2, \dots, \beta_J)$ by the requirement that $f(z)$ integrate to 1 over the range of z . The choice $J = 2$ makes (5.10) into the normal family $\mathcal{N}(\mu, \sigma^2)$.

Exercise 5.2 Write the $\mathcal{N}(\mu, \sigma^2)$ family in the form (5.10) showing the expressions for β_0, β_1 , and β_2 .

¹ Appendix A gives a brief review of exponential family theory, including Lindsey's method as used in what follows.

If we wish to detect differences between $f(z)$ and the theoretical null density $f_0(z) = \varphi(z)$, we will certainly need J bigger than 2. As J goes to infinity, family (5.10) grows to approximate all densities on the line. This is the non-parametric ideal, but not an efficient choice when we expect $f(z)$ to be quite smooth. The program `locfdr` discussed below defaults to $J = 7$. Relying on family (5.10) with $J = 7$ puts us, very roughly, midway between traditional parametric and non-parametric estimation.

Maximum likelihood estimation in family (5.10) seems like it would require special software. This isn't the case. *Lindsey's method*, an algorithm based on discretizing the z_i values, obtains maximum likelihood estimates $\hat{\beta}$ using standard Poisson regression techniques. We partition the range $\mathcal{Z}$ of the z_i values into K bins of equal width d ,

$$\mathcal{Z} = \bigcup_{k=1}^K \mathcal{Z}_k. \quad (5.11)$$

The histograms in Figure 5.1a used $K = 90$ bins of width $d = 0.1$ over range $\mathcal{Z} = [-4.5, 4.5]$. Define y_k as the count in the k th bin,

$$y_k = \#\{z_i \in \mathcal{Z}_k\} \quad (5.12)$$

and let

$$x_k = \text{centerpoint of } \mathcal{Z}_k \quad (5.13)$$

so $x_1 = -4.45$, $x_2 = -4.35, \dots, x_{90} = 4.45$ in Figure 5.1a. The expected value of y_k is approximately

$$\nu_k = Nd f(x_k) \quad (5.14)$$

where N is the total number of cases, respectively 6033 and 15443 in Figures 5.1a and 5.1b. Lindsey's method assumes that the y_k are independent Poisson counts

$$y_k \stackrel{\text{ind}}{\sim} \text{Poi}(\nu_k) \quad k = 1, 2, \dots, K, \quad (5.15)$$

and then fits $\hat{f}$ via a regression model for ν_k as a function of x_k . In model (5.10),

$$\log(\nu_k) = \sum_{j=0}^J \beta_j x_k^j; \quad (5.16)$$

(5.15)–(5.16) is a standard Poisson generalized linear model (GLM). Program `locfdr`² is written in the language R which permits a one-line call

² Appendix B describes `locfdr`.

for the MLE $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_J)$. Except for the effects of discretization, which are usually negligible, $\hat{\beta}$ is identical to the MLE we would obtain by direct maximization of $\prod_1^N f(z_i)$ in (5.10), assuming independence.

Lindsey's method has the nice effect of moving density estimation into the more familiar realm of regression: we are fitting a smooth function $f_{\beta}(x_k)$ to the counts y_k . (Technically, this is done by minimizing the sum of Poisson deviances,

$$\sum_{k=1}^K D(y_k, Ndf_{\beta}(x_k)), \quad D(y, v) = 2y \left[\log\left(\frac{y}{v}\right) - \left(1 - \frac{y}{v}\right) \right], \quad (5.17)$$

over the choice of β .) Independence of the z_i or of the y_k in (5.15) is not required; $\hat{f}$ tends to be consistent and close to unbiased for f even under dependence, the penalty for dependence being increased variability, as described in Chapter 7. The fit can be examined by eye or more formally, and adjusted as necessary. Form (5.10) for $f(x)$ isn't required. By default, `locfdr` replaces the polynomial exponent with a natural spline basis. Doing so had very little effect on Figure 5.1a and Figure 5.2, nor did reducing J from 7 to 5.

Our interest in $\hat{f}(z)$ arises from its use in the denominator of $\widehat{\text{fdr}}(z) = \hat{\pi}_0 f_0(z) / \hat{f}(z)$. The examples in Figure 5.2 are typical: $\widehat{\text{fdr}}(z)$ is about 1 near $z = 0$ and either does or does not decline toward zero in the tails. The transition points from high to low $\widehat{\text{fdr}}$, denoted (following criterion (5.7)) by the small triangles in Figure 5.2, claim our attention. It is near such points where the accuracy of $\hat{f}(z)$ is most important.

It seems, intuitively, that $\widehat{\text{fdr}}$ should be harder to estimate than Fdr , but that is not the case, at least not within the purview of models like (5.10). Figure 5.3 compares the standard deviation of $\log(\widehat{\text{fdr}}(z))$ with those of $\log(\widehat{\text{Fdr}}(z))$ and $\log(\overline{\text{Fdr}}(z))$. The situation involves $N = 6000$ independent observations from the two-groups model (2.7),

$$\begin{aligned} \pi_0 &= 0.95 & z &\sim \mathcal{N}(0, 1) \\ \pi_1 &= 0.05 & z &\sim \mathcal{N}(2.5, 1); \end{aligned} \quad (5.18)$$

$\widehat{\text{fdr}}(z)$, $\widehat{\text{Fdr}}(z)$, and $\overline{\text{Fdr}}(z)$ were estimated as in (5.5), (5.6), and Section 4.3, except that $\pi_0 = 0.95$ was assumed known in all three cases. Both $\widehat{\text{fdr}}$ and $\widehat{\text{Fdr}}$ used a natural spline basis with $J = 5$ degrees of freedom. Their standard deviations were computed from the theoretical formulas in Chapter 7, and checked with simulations.

In this situation, $\widehat{\text{fdr}}(z)$ is seen to be a little *less* variable than $\widehat{\text{Fdr}}(z)$, which in turn is less variable than the non-parametric version $\overline{\text{Fdr}}(z)$. All

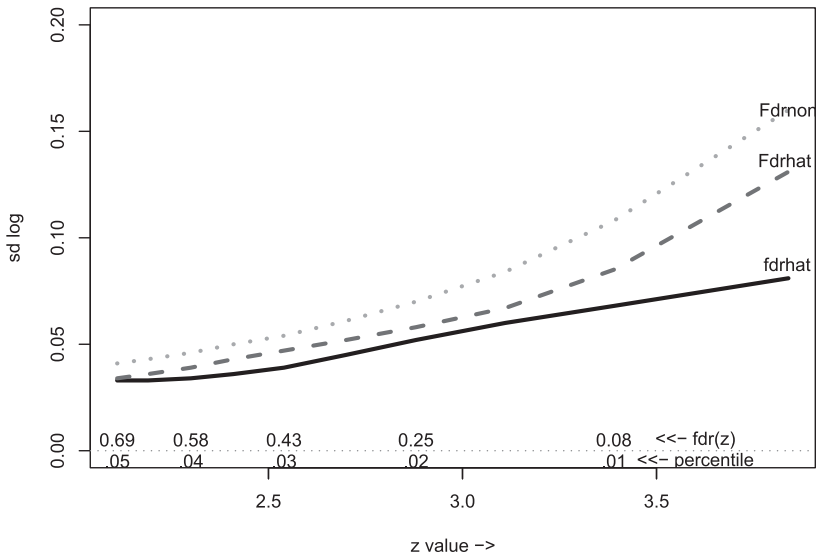


Figure 5.3 Comparison of standard deviations for $\log(\widehat{\text{fdr}}(z))$, $\log(\widehat{\text{Fdr}}(z))$, and $\log(\overline{\text{Fdr}}(z))$ for $N = 6000$ independent observations from model (5.18). Small numbers show true values of $\text{fdr}(z)$ at the indicated right percentiles of the mixture distribution. Standard deviations are based on the results in Chapter 7.

three estimates are satisfyingly accurate: at the 0.01 right percentile of the mixture distribution, $\text{sd}\{\log \widehat{\text{fdr}}(z)\} = 0.068$, implying that $\widehat{\text{fdr}}(z)$ has coefficient of variation only about 7%. Correlation among the z_i values makes the standard deviations worse, as demonstrated in Chapter 7, but the overall comparison still shows the three estimates to be of roughly equal variability, with an edge to $\widehat{\text{fdr}}(z)$ in the extreme tails.

Exercise 5.3 If X is a positive random variable with mean μ and variance σ^2 , show that $\text{CV} \equiv \sigma/\mu$, the *coefficient of variation* of X , approximately equals $\text{sd}\{\log(X)\}$ when both are small.

5.3 Inference and Local False Discovery Rates

Switching attention from Fdr to fdr moves us still closer to Bayesian estimation theory: it is more appropriate from the Bayesian point of view to estimate $\text{fdr}(z) = \Pr\{\text{null}|z\}$ than the tail-area version $\Pr\{\text{null}|Z \geq z\}$.

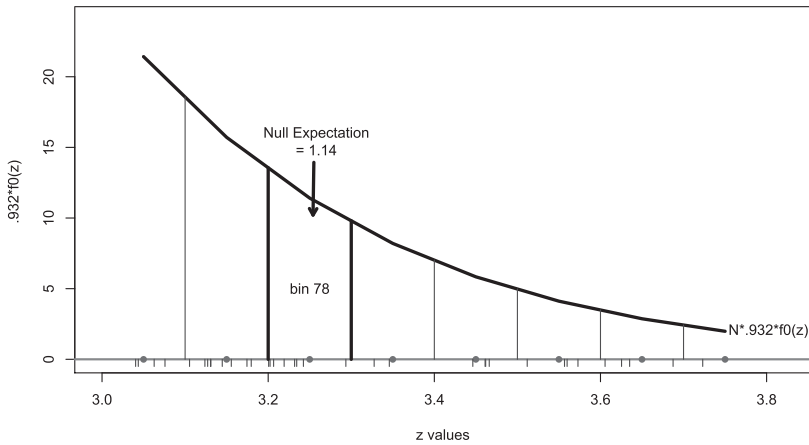


Figure 5.4 Right tail of the prostate data histogram, Figure 5.1a. Heavy curve is estimated $\widehat{\text{fdr}}$ numerator $N \cdot 0.932 \cdot f_0(z)$ (5.5). Hash marks indicate observed values z_i . Bin 78 contains 8 z_i values and has null expectation (area under the curve) of 1.14, giving raw estimate $\widehat{\text{fdr}}_{78} = 1.14/8 = 0.143$. Smoothing the counts using $\hat{f}(z)$ gave $\widehat{\text{fdr}}_{78} = 0.254$.

Figure 5.4 uses the prostate data to illustrate the simple idea behind the estimate $\widehat{\text{fdr}}(z)$. Histogram bin $k = 78$, $z \in [3.2, 3.3)$, is observed to contain $y_k = 8$ z -values; the expected null count is 1.14, leading to the raw estimate

$$\widehat{\text{fdr}}_{78} = 1.14/8 = 0.143. \quad (5.19)$$

However, the counts y_k are highly variable. Replacing them with smooth values $\hat{y}_k = N \cdot d \cdot \hat{f}(x_k)$, equaling 4.49 for $k = 78$, gave

$$\widehat{\text{fdr}}_{78} = 1.14/4.49 = 0.254. \quad (5.20)$$

Table 5.1 provides data for bins 76 through 83, emphasizing the importance of smoothing.

The use of local false discovery rates raises several issues concerning both technique and inference.

- *More general structure* The two-groups model (5.1) can be generalized to allow the cases $i = 1, 2, \dots, N$ to behave differently:

$$\begin{aligned} \pi_{i0} &= \Pr\{\text{case } i \text{ null}\} & f_{i0}(z) &= \text{case } i \text{ null density} \\ \pi_{i1} &= \Pr\{\text{case } i \text{ non-null}\} & f_{i1}(z) &= \text{case } i \text{ non-null density.} \end{aligned} \quad (5.21)$$

Table 5.1 Data for bins $k = 76$ through 83, prostate data, Figure 5.4; e_{0k} is null expected count (area under the curve); y_k is observed count, $\hat{y}_k$ smoothed version $N \cdot d \cdot \hat{f}(x_k)$; $\overline{\text{fdr}}_k = e_{0k}/y_k$, $\widehat{\text{fdr}}_k = e_{0k}/\hat{y}_k$.

k	x_k	e_{0k}	y_k	$\overline{\text{fdr}}_k$	$\hat{y}_k$	$\widehat{\text{fdr}}_k$
76	3.05	2.14	4	.536	5.96	.359
77	3.15	1.57	9	.175	5.14	.306
78	3.25	1.14	8	.143	4.49	.254
79	3.35	.82	2	.410	3.97	.207
80	3.45	.58	4	.146	3.55	.165
81	3.55	.41	4	.103	3.20	.128
82	3.65	.29	4	.072	2.92	.098
83	3.75	.20	1	.198	2.69	.074

Define

$$\begin{aligned}\pi_0 &= \frac{1}{N} \sum_{i=1}^N \pi_{i0} & f_0(z) &= \frac{1}{N} \sum_{i=1}^N \frac{\pi_{i0}}{\pi_0} f_{i0}(z) \\ \pi_1 &= \frac{1}{N} \sum_{i=1}^N \pi_{i1} & f_1(z) &= \frac{1}{N} \sum_{i=1}^N \frac{\pi_{i1}}{\pi_1} f_{i1}(z).\end{aligned}\tag{5.22}$$

Marginally (that is, averaging over $i = 1, 2, \dots, N$) we are back in the two-groups situation, with $\pi_0, \pi_1, f_0(z), f_1(z)$, and $f(z) = \pi_0 f_0(z) + \pi_1 f_1(z)$ having the same meanings as in (5.1) and Figure 2.3, and with $\text{fdr}(z) = \pi_0 f_0(z)/f(z) = \Pr\{\text{null}|z\}$ (5.2).

Exercise 5.4 (i) Verify the statements above. (ii) Why is $\hat{f}(z)$, obtained as in Section 5.2, a valid estimate for $f(z)$ under model (5.21)? *Hint:* (5.14).

These results can be visualized in terms of Figure 5.4. Under model (5.21) the expected number of null cases falling into bin k is

$$e_0(k) \doteq d \sum_{i=1}^N \pi_{i0} f_{i0}(x_k) = Nd \pi_0 f_0(x_k),\tag{5.23}$$

d = bin width, the same approximation we would use for model (5.1). The estimate $\widehat{\text{fdr}}(z) = \hat{\pi}_0 f_0(z)/\hat{f}(z)$ is obtained entirely from the empirical distribution of the $N z_i$ values so model (5.21) gives the same results as (5.1), at least as long as we ignore distinctions among the cases. An assumption such as $f_0(z) \sim \mathcal{N}(0, 1)$ is still valid if it applies to the *average* null case,

not necessarily to each of them. These same comments hold true for $\widehat{\text{Fdr}}(z)$ and $\widehat{\text{Fdr}}(z_i)$.

- *Using prior knowledge* All of this assumes that model (5.21) is invisible to us and that we can only work marginally with the z_i values. If (5.21) were known, then

$$\text{fdr}_i(z_i) = \pi_{i0} f_{i0}(z_i) / f_i(z_i) = \Pr\{\text{case } i \text{ null} | z_i\} \quad (5.24)$$

would be more relevant than $\text{fdr}(z_i) = \pi_0 f_0(z_i) / f(z_i)$.

Exercise 5.5 Suppose that in model (5.21) $f_{i0}(z)$ and $f_i(z)$ do not depend on i . Show that

$$\text{fdr}_i(z_i) = \text{fdr}(z_i) \frac{r_i}{1 - (1 - r_i) \text{fdr}(z_i)} \quad \text{when } r_i = \frac{\pi_{i0}/(1 - \pi_{i0})}{\pi_0/(1 - \pi_0)}. \quad (5.25)$$

Gene 637 of the prostate data, Table 4.3, had $\widehat{\text{fdr}}(z_{637}) = 0.105$. If gene 637 was on a short list of “hot prospects” provided a priori by the investigator, we could assume $r_{637} < 1$ (i.e., that $\pi_{637,0}$ is less than the average π_{i0}), perhaps taking $r_{637} = 0.5$, in which case (5.25) would yield $\widehat{\text{fdr}}_{637}(z_{637}) = 0.055$. Chapter 10 concerns working with models like (5.21) in situations where covariate information is available to assist with the estimation of $\text{fdr}_i(z_i)$.

- *Exchangeability* The right-sided tail area $\widehat{\text{Fdr}}$ for the prostate data has $\widehat{\text{Fdr}}(3.2) = 0.108$. We might report the 36 genes having $z_i \geq 3.2$ as interesting prospects for further prostate cancer investigation, since fewer than four of them are expected to be null. A tacit exchangeability assumption is at work here: each of the 36 is implied to have probability about 1/9 of being null. This ignores the fact that some of the 36 z_i 's are much greater than others, ranging from 3.20 to 5.29.

The local fdr puts less strain on exchangeability. We can interpret $\widehat{\text{fdr}}_k$ in Table 5.1 as saying that the eight cases with z_i in $[3.2, 3.3)$ each have probability 0.25 of being null, the two cases in $[3.3, 3.4)$ each have probability 0.21, etc. Of course, exchangeability is lost if we have different prior knowledge about the cases, say, different values of r_i in (5.25). There we used Bayes theorem to appropriately adjust $\text{fdr}(z_i)$.

- *Scaling properties* $\widehat{\text{Fdr}}$ and fdr methods scale in a nice way as the number N of cases changes. Suppose case 1 of the prostate data had $z_1 = 3.25$, giving $\widehat{\text{fdr}}(z_1) = 0.25$. What would $\widehat{\text{fdr}}(3.25)$ be if the investigator had

measured twice as many genes? The answer is probably about the same, at least if we believe model (5.1): doubling N will give more accurate estimates of $f(z)$ and π_0 but major changes in $\hat{\pi}_0 f_0(z)/\hat{f}(z)$ are unlikely. Even letting N go to infinity shouldn't greatly change $\widehat{\text{fdr}}(z)$, as it converges to $\text{fdr}(z) = \pi_0 f_0(z)/f(z)$ in the usual asymptotic manner.

In contrast, inferences from the FWER methods of Chapter 3 depend crucially on the number of cases N . The right-sided p -value for $z_1 = 3.25$ is

$$p_1 = 1 - \Phi(3.25) = 0.00058 \quad (5.26)$$

under the theoretical null $\mathcal{N}(0, 1)$ distribution. Bonferroni's method (3.13) would declare case 1 non-null at significance level α if

$$p_1 \leq \alpha/N, \quad (5.27)$$

so doubling N lowers by half the threshold for rejection; p_1 would be *significant* if N were $< \alpha/p_1$ but not otherwise, the significance condition in (5.26) being $N \leq 86$ for $\alpha = 0.05$.

Exercise 5.6 Using the two-groups model, show that as $n \rightarrow \infty$, Holm's procedure (3.20) requires, asymptotically,

$$p_1 \leq \frac{\alpha}{N(1 - P_1)} \quad \text{with } P_1 = 1 - F(z_1) \quad (5.28)$$

as a necessary but not sufficient condition for rejection. (For Hochberg's step-up procedure, Exercise 3.8, condition (5.28) is sufficient but not necessary.)

Let $z_{(1)}$ be the smallest (most negative) of the z_i values, with corresponding p -value $p_{(1)} = F_0(z_{(1)})$. Then $\widehat{\text{Fdr}}(z_{(1)}) = \pi_0 F_0(z_{(1)})/\bar{F}(z_{(1)})$ equals $N\pi_0 p_{(1)}$. This will be less than the control level q if

$$p_{(1)} \leq \frac{1}{N} \frac{q}{\pi_0}, \quad (5.29)$$

which is *the same as the Bonferroni bound* with $\alpha = q/\pi_0$. In other words, the nice scaling properties of false discovery rate methods break down for extreme inferences. We will see this happening in the "snp" example of Chapter 11.

Another limitation concerns *relevance*. Model (5.1) taken literally implies an infinite reservoir of relevant cases. In practice though, if we doubled the number of genes in the prostate study, the new ones — not previously included on the microarray roster — might be less relevant to in-

ferences about gene 1. Chapter 10 concerns the relevance question, which applies to all multiple inference procedures, not just false discovery rates.

- *Models with more structure* Exponential family models like (5.10) can be elaborated to handle situations with more structure than the two-groups model (5.11). Suppose the N cases are naturally partitioned into M classes, perhaps representing different regions of the brain in the DTI example. We could apply `locfdr` separately to each class, but this invites estimation problems in the smaller classes. A more efficient approach expands (5.10) to

$$\log\{f(z)\} = \sum_{j=0}^J \beta_j z^j + \gamma_{1m} z + \gamma_{2m} z^2, \quad (5.30)$$

m indicating the class, with $\sum_m \gamma_{1m} = \sum_m \gamma_{2m} = 0$, effectively allowing different means and variances for $\hat{f}$ in each group while retaining common tail behavior. Similar methods are discussed in Chapter 10.

- *Combining Fdr and fdr* It is not necessary to choose between $\widehat{Fdr}$ and $\widehat{fdr}$; they can be used in tandem. For example, $\widehat{Fdr}(3.20) = 0.108$ for the prostate data, right-sided, applies to all 36 z_i 's exceeding 3.20 while $\widehat{fdr}(3.25) = 0.254$ applies to those in $[3.2, 3.3)$, etc., as suggested earlier, giving the investigator both a list of likely prospects and quantitative differentiation within the list. Program `locfdr` automatically puts $\widehat{Fdr}$ and $\widehat{fdr}$ in relationship (2.15),

$$\widehat{Fdr}(z) = \int_{-\infty}^z \widehat{fdr}(x) \hat{f}(x) dx \Bigg/ \int_{-\infty}^z \hat{f}(x) dx, \quad (5.31)$$

and similarly for the right-sided $\widehat{Fdr}$.

- *Bayes limitation* The empirical Bayes inference that $\widehat{fdr}(z_i)$ estimates $\Pr\{\text{case } i \text{ null} | z_i\}$ (5.2) comes with a caveat: it is not necessarily an estimate of

$$\Pr\{\text{case } i \text{ null} | z_1, z_2, \dots, z_N\}; \quad (5.32)$$

(5.32) can be different than (5.2) if the z -values are correlated. Chapter 9 on *enrichment* concerns methods for making inferences on groups of cases, rather than one at a time.

- *Expected false and true positives* Local false discovery rates are directly related to “EFP” and “ETP”, the expected number of false positives

and true positives of a decision procedure. Suppose that z_i follows model (5.21) for $i = 1, 2, \dots, N$ and that we intend to reject the i th null hypothesis for $z_i \geq c_i$. The size and power of the i th test are

$$\alpha_i = \int_{c_i}^{\infty} f_{i0}(z_i) dz_i \quad \text{and} \quad \beta_i = \int_{c_i}^{\infty} f_{i1}(z_i) dz_i \quad (5.33)$$

giving

$$\text{EFP} = \sum_{i=1}^N w_i \pi_{i0} \alpha_i \quad \text{and} \quad \text{ETP} = \sum_{i=1}^N w_i \pi_{i1} \beta_i \quad (5.34)$$

where w_i is the prior probability of case i (which we can take to be $1/N$ without affecting the following calculations).

We wish to maximize ETP for a given value of EFP, by an optimum choice of the cutoff values $\mathbf{c} = (c_1, c_2, \dots, c_N)$. Since

$$\frac{\partial \text{EFP}}{\partial c_i} = \sum_{i=1}^N w_i \pi_{i0} \frac{\partial \alpha_i}{\partial c_i} = - \sum_{i=1}^N w_i \pi_{i0} f_{i0}(c_i) \quad (5.35)$$

and similarly $\partial \text{ETP} / \partial c_i = - \sum w_i \pi_{i1} f_{i1}(c_i)$, a standard Lagrange multiplier argument shows that at the optimal value of $\mathbf{c}$,

$$\pi_{i1} f_{i1}(c_i) = \lambda \pi_{i0} f_{i0}(c_i) \quad \text{for } i = 1, 2, \dots, N \quad (5.36)$$

for some constant λ . In terms of (5.24) this is equivalent to

$$\text{fdr}_i(c_i) = 1/(1 + \lambda). \quad (5.37)$$

That is, the decision rule that maximizes ETP for a given value of EFP *rejects each null hypothesis at the same threshold value of its local false discovery rate*. If all the cases follow the same two-groups model (5.1), then the optimum rule rejects for large values of $\text{fdr}(z_i)$ (5.2).

5.4 Power Diagnostics

The FWER techniques of Chapter 3 are aimed at the control of Type I errors, the rejection of null hypotheses that are actually correct. Discussion of false discovery rates, as the name suggests, similarly tends to focus on a form of Type I error control. However, Fdr theory also has something useful to say about *power*, the probability of rejecting null hypotheses that *should be* rejected. This section discusses some simple power diagnostics based on local false discovery rates.

Starting from the two-groups model (5.1), define the *local true discovery rate* $\text{tdr}(z)$ to be

$$\text{tdr}(z) = \Pr\{\text{non-null}|z\} = 1 - \text{fdr}(z) = \pi_1 f_1(z)/f(z), \quad (5.38)$$

the last equality following from (5.2)–(5.3). An fdr estimate $\widehat{\text{fdr}}(z) = \hat{\pi}_0 \hat{f}_0(z)/\hat{f}(z)$ also gives a tdr estimate

$$\widehat{\text{tdr}}(z) = 1 - \widehat{\text{fdr}}(z) = \hat{\pi}_1 \hat{f}_1(z)/\hat{f}(z) \quad (5.39)$$

where

$$\hat{\pi}_1 = 1 - \hat{\pi}_0 \quad \text{and} \quad \hat{f}_1(z) = \frac{\hat{f}(z) - \hat{\pi}_0 \hat{f}_0(z)}{1 - \hat{\pi}_0}. \quad (5.40)$$

Exercise 5.7 Verify (5.38) and (5.40).

Count y_k (5.12) includes all cases in bin k , null or non-null. We cannot, of course, separate y_k into nulls and non-nulls, but we can compute an approximation for the *expected* non-null count,

$$y_{1k} = \widehat{\text{tdr}}_k \cdot y_k \quad \left[\widehat{\text{tdr}}_k = \widehat{\text{tdr}}(x_k) \right]. \quad (5.41)$$

The rationale for (5.41) is based on the fact that tdr_k is the proportion of non-null cases in bin k .

Exercise 5.8 Assuming the two-groups model (5.1) and independence among the z_i , what is the distribution of the number of nulls in bin k , given y_k ? How does this support definition (5.41)?

The non-null counts y_{1k} can suffer from excess variability due to “histogram noise” in the observed counts y_k . An improved version is

$$\hat{y}_{1k} = Nd \widehat{\text{tdr}}_k \cdot \hat{f}_k \quad \left[\hat{f}_k = \hat{f}(x_k) \right] \quad (5.42)$$

as in (5.14), called the *smoothed non-null counts* in what follows.

The vertical bars in Figure 5.5 are non-null counts y_{1k} for the prostate data using the histogram bins pictured in Figure 5.1a. The solid curve traces the smoothed non-null counts y_{1k} . There is one important difference here from the previous discussion. Rather than taking $f_0(z)$ to be the $\mathcal{N}(0, 1)$ theoretical null (2.9), the estimate of $\text{fdr}(z) = \pi_0 f_0(z)/f(z)$ used

$$\hat{f}_0(z) \sim \mathcal{N}(0, 1.06^2); \quad (5.43)$$

(5.43) is the *empirical null* discussed in Chapter 6, the normal distribution that best fits the z -value histogram near its center. This change increased $\hat{\pi}_0$ from 0.932 in (5.4) to

$$\hat{\pi}_0 = 0.984. \quad (5.44)$$

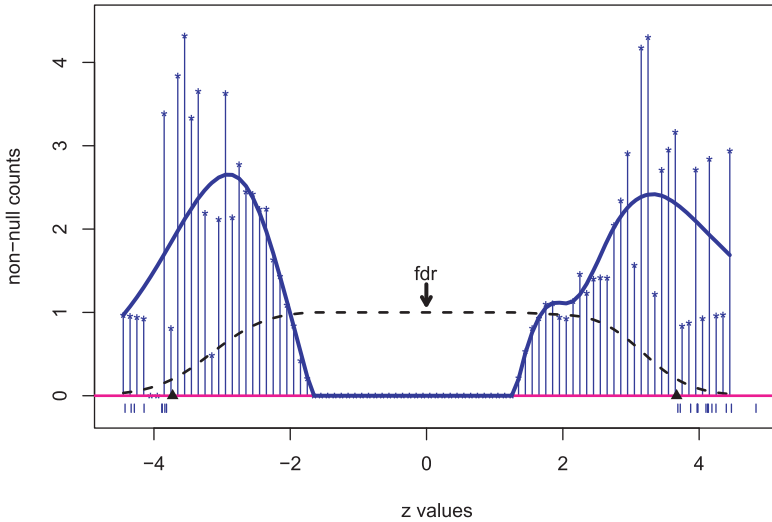


Figure 5.5 Vertical bars indicate non-null counts y_{1k} for the prostate data, taking $\hat{\pi}_0 = 0.984$ and $\hat{f}_0 \sim \mathcal{N}(0, 1.06^2)$; histogram bins as in (5.12)–(5.13). Heavy curve follows smoothed non-null counts (5.42); dashed curve is $\widehat{\text{fdr}}(z)$. Twenty-three of the z_i have $\widehat{\text{fdr}}(z_i) \leq 0.20$, indicated by small hash marks. Most of the non-null counts occur when $\widehat{\text{fdr}}(z)$ is high, indicating low power.

The difference between (5.43) and $f_0 \sim \mathcal{N}(0, 1)$ becomes crucial in power calculations, which in general are more delicate than questions relating to Type I error.

The estimate $\widehat{\text{fdr}}(z)$ (broken curve) is less optimistic than its counterpart in Figure 5.2. Now only 23 genes, those with z_i less than -3.73 or greater than 3.67 , have $\widehat{\text{fdr}}(z_i) \leq 0.20$. The total smoothed non-null count $\sum \hat{y}_{1k}$ equals 105, which is a little greater than $(1 - \hat{\pi}_0) \cdot N = 97$ because some small negative values near $z = 0$ have been suppressed.

Exercise 5.9 Show that $\int_{-\infty}^{\infty} \text{tdr}(z)f(z)dz = \pi_1$. Why does this suggest $\sum_k y_{1k} \doteq (1 - \hat{\pi}_0) \cdot N$?

Of the 105 total smoothed non-null counts, only 26.8 occur in the regions where $\widehat{\text{fdr}}(z) \leq 0.20$, about 26%. The bulk occur where $\widehat{\text{fdr}}(z)$ is high, more than half with $\widehat{\text{fdr}}(z) > 0.50$. So if we try to report more than the list of 23 genes as interesting prospects, we soon expose the investigators to high

probabilities of disappointment. In other words, the prostate study has low power.

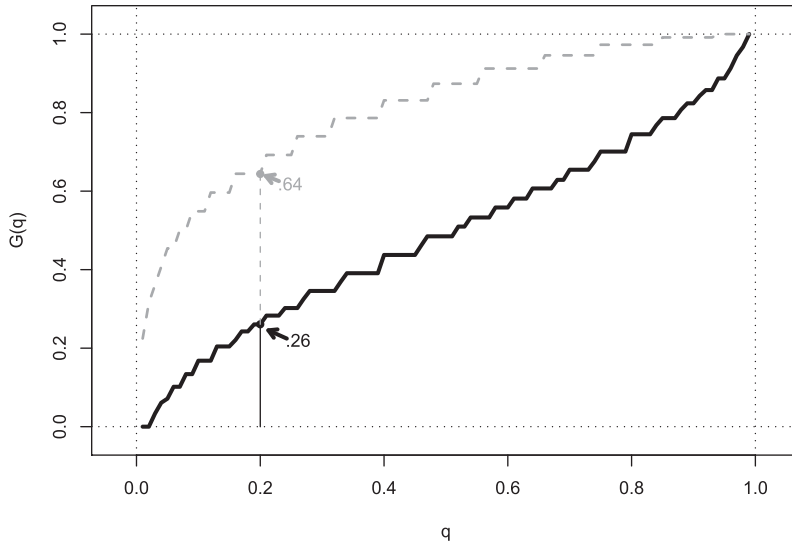


Figure 5.6 Heavy curve is smoothed non-null cdf $\hat{G}(q)$ for the estimated local false discovery rate; for example, $\hat{G}(0.2) = 0.26$ indicates that only 26% of the $\hat{y}_{1k}$ values occurred where $\widehat{\text{fdr}}_k \leq 0.20$. Dashed curve is $\hat{G}(q)$ for a simulated sample of $N = 6000$ independent z_i values, 95% from $\mathcal{N}(0, 1)$ and 5% from $\mathcal{N}(3, 1)$. Here 64% of the $\hat{y}_{1k}$ values had $\widehat{\text{fdr}}_k \leq 0.20$.

Figure 5.6 shows the entire non-null cdf of $\widehat{\text{fdr}}(z)$,

$$\Pr_1 \{ \widehat{\text{fdr}} \leq q \} = \sum_{k: \widehat{\text{fdr}}_k \leq q} \hat{y}_{1k} / \sum_k \hat{y}_{1k} \quad [q \in (0, 1)]. \quad (5.45)$$

For example, $G(0.2) = 0.26$ and $G(0.5) = 0.49$. A simulated high-powered situation is shown for comparison, in which there are $N = 6000$ independent z_i 's, $\pi_0 = 95\%$ having $z_i \sim \mathcal{N}(0, 1)$ and $\pi_1 = 5\%$ with $z_i \sim \mathcal{N}(3, 1)$. It had $\hat{G}(0.2) = 0.64$. In a typical realization of this situation, a majority of the non-null cases would be reported on the list of those with $\widehat{\text{fdr}}(z_i) \leq 0.20$, compared with about one-quarter for the prostate study.

A simple but useful diagnostic summary statistic³ for power is $\widehat{E \text{fdr}}_1$,

³ Program `locfdr` returns $\widehat{E \text{fdr}}_1$ and also separate versions of (5.46) applying to the left and right tails.

the expectation of $\widehat{\text{fdr}}_k$ under the $\hat{y}_{1k}$ distribution,

$$\widehat{E \text{fdr}}_1 = \sum_k \widehat{\text{fdr}}_k \hat{y}_{1k} \bigg/ \sum_k \hat{y}_{1k}. \quad (5.46)$$

Low values of $\widehat{E \text{fdr}}_1$ indicate high power — that a typical non-null case will occur where the local false discovery rate is low — and high values indicate low power. For the prostate study, $\widehat{E \text{fdr}}_1 = 0.51$, indicating that a typical non-null case could expect to have $\widehat{\text{fdr}}$ about 0.5. For the simulated sample, $\widehat{E \text{fdr}}_1 = 0.28$.

Table 5.2 *Simulation study of $\widehat{E \text{fdr}}_1$, power diagnostic (5.46). Model (5.1), $\pi_0 = 0.95$, $f_0 \sim \mathcal{N}(0, 1)$, and $\pi_1 = 0.05$, $f_1 \sim \mathcal{N}(3, 1)$; number of cases $N = 1500, 3000$, or 6000 . Boldface shows mean over 100 simulations, with standard deviations in parentheses. True $E \text{fdr}_1 = 0.329$.*

N	$\widehat{E \text{fdr}}_1$	$\hat{\pi}_0$
1500	.266 (.046)	.965 (.008)
3000	.272 (.038)	.963 (.007)
6000	.281 (.028)	.962 (.005)

Table 5.2 reports on a small simulation study of $\widehat{E \text{fdr}}_1$. As in Figure 5.6, each simulation had N independent z_i values, $\pi_0 = 0.95$ of them $\mathcal{N}(0, 1)$ and $\pi_1 = 0.05$ from $\mathcal{N}(3, 1)$, $N = 1500, 3000$, or 6000 . The true value is $E \text{fdr}_1 = 0.329$. We see that $\widehat{E \text{fdr}}_1$ is quite stable but biased downward. Most of the bias can be traced to the upward bias of $\hat{\pi}_0$, which reduces $\widehat{\text{tdr}}(z)$ (5.39). Since

$$\widehat{E \text{fdr}}_1 = \sum_k \widehat{\text{tdr}}_k \cdot \widehat{\text{fdr}}_k \cdot \hat{f}_k \bigg/ \sum_k \widehat{\text{fdr}}_k \cdot \hat{f}_k, \quad (5.47)$$

this leads to underestimates of $\widehat{E \text{fdr}}_1$. Tracing things back further, the upward bias of $\hat{\pi}_0$ arises from taking literally the zero assumption (4.44).

Exercise 5.10 Numerically verify the “true value” $E \text{fdr}_1 = 0.329$.

Investigators often object that their pre-experimental favorites have not appeared on the reported $\widehat{\text{fdr}} \leq 0.20$ (or $\overline{\text{Fdr}} \leq 0.10$) list. Low power is a likely culprit. If the prostate study were rerun from scratch, another list of 23 or so genes might be reported, barely overlapping with the original list.

Z-values, even ones from genuinely non-null cases, are quite variable (see Section 7.4), and in low-power situations a case has to be lucky as well as good to be reported. The Bayesian formula (5.25) may help salvage some of the investigators' hot prospects.

Finally, it should be noted that all of these power diagnostics are computed from the data at hand, the original set of z -values, and do not require prior knowledge of the situation. This is one of the advantages of large-scale studies.

Notes

Local false discovery rates were introduced as a natural extension of tail area Fdrs in Efron et al. (2001). Techniques both more and less parametric than Lindsey's method have been proposed for estimating $f(z)$ and $\text{fdr}(z)$, all of which seem to perform reasonably well: Allison et al. (2002), Pan et al. (2003), Pounds and Morris (2003), Heller and Qing (2003), Broberg (2004), Aubert et al. (2004) and Liao et al. (2004). Recent methods, which also encompass Chapter 6's problem of estimating the null density, include Jin and Cai (2007) and Muralidharan (2010), the former using Fourier series and the latter based on normal mixture models.

Estimation of the mixture density $f(z)$ was a problem of considerable interest in the period following Robbins' path-breaking empirical Bayes papers. That literature was more mathematics-oriented than our current computation-intensive efforts. A nice summary and example is found in Singh (1979), where the asymptotics of kernel estimates for $f(z)$ are carefully elaborated.

Power diagnostics are discussed in Section 3 of Efron (2007b). A familiar power-related question is also considered there: how much increased power could we obtain by increasing the number n of subjects? Answers to such a question are necessarily speculative, but simple techniques provide at least a rough guide. For the prostate data they suggest that tripling n would halve $E \text{fdr}_1$.

It isn't necessary that the individual summary statistics z_i be scalars. Tibshirani and Efron (2002) analyze an example in which each z_i is two-dimensional. The null density f_0 is more difficult to define and estimate in this case, but the fdr definition (5.2) still makes sense.

Storey (2007) suggests maximizing ETP as a function of EFP. His "optimum discovery procedure" proposes a different rule than (5.37), applicable to a restricted class of testing procedures.